An Investigation of Salt Toxicity

Overview and Purpose

This lab is supposed to help us understand that human activities can change the lives of other organisms because of their sensitive nature, they need a very specific range in their environment to survive. In this lab, we will learn how to find the upper limit of a range of tolerance for a pollutant on a plant species, create a dose response curve, and determine the LD_{50} graphically for a chemical.

PRE-LAB QUESTIONS

1. Describe what happens to road salt after it is applied to roads.

Road salt lowers the freezing point of water so if there is snow or ice on the road, they melt at a temperature they normally wouldn't.

2. Describe two impacts of increased salt concentration on freshwater organisms.

Increased salt concentration can be toxic for freshwater organisms and a higher concentration of salt in freshwater can cause dehydration in those organisms because of osmosis.

3. Explain how plant life can be impacted by increased salt concentrations.

Plant life can also be affected by increased salt concentrations as they will become dehydrated because of osmosis.

Lab Investigation

PROCEDURE

Day 1

1. Your teacher will assign you to a group and provide your group with a 3 percent stock solution of one or more of the following: NaCl, $MgCl_2$, or $CaCl_2$.
2. Using the stock solution, prepare 10 mL of each of the following solutions. 3.0% / 2.5% / 2.0% / 1.5% / 1.0% / 0.5% / 0.1% / 0.05% / 0.01% / 0.0% (control)

   Example: To prepare 10 mL of a 2.5 percent solution, add 8.3 mL of 3 percent stock solution to a graduated cylinder, add distilled water to a total volume of10 mL, and pipette up and down or use a stirring rod to mix.

$$(3\%)(V_I) = (2.5\%)(10\ \text{mL})$$

$$V_I = \frac{(2.5\%)(10\ \text{mL})}{3\%} = 8.3\ \text{mL}$$

3. Label 10 petri dishes with the salt concentrations and your group number. Nine will be test solutions, and one will be a control (pure distilled water).
4. Cut circles out of paper towels to fit the bottom of the petri dishes, and place one circle in each dish. Using a beral pipet, transfer just enough of the 3 percent solution to saturate the paper towel in the appropriately labeled Petri dish. Repeat this for the remaining nine dishes, matching the solution used to the dish label and making sure to add the same amount of solution each time.
5. Place 10 radish seeds, evenly spaced, into each petri dish.
6. Stack the petri dishes and store them as directed by your teacher.

Research Design Component	**In This Experiment**
Independent variable	% Salt Concentration
Dependent variable	% Mortality in Seeds
Control	0% concentration of salt
Constant	Amount of seeds

Day 2

1. After the seeds have germinated for 3–4 days, it is time to collect the data.

Concentration (% salt)	# of seeds that did not germinate	Average root length (mm)	# of seeds that did not germinate (class average)	% Mortality (class average)	Class average root length (mm)
3	10	0.00	50	100%	0.00
2.5	10	0.00	50	100%	0.00
2.0	10	0.00	43	86%	2.98
1.5	9	1.50	37	74%	10.58
1.0	2	7.30	11	22%	15.36
0.5	0	20.40	0	0%	24.99
0.1	0	33.10	11	22%	32.64
0.05	2	17.50	3	6%	32.98
0.01	0	35.20	3	6%	36.14
0.0	1	20.50	1	2%	41.6

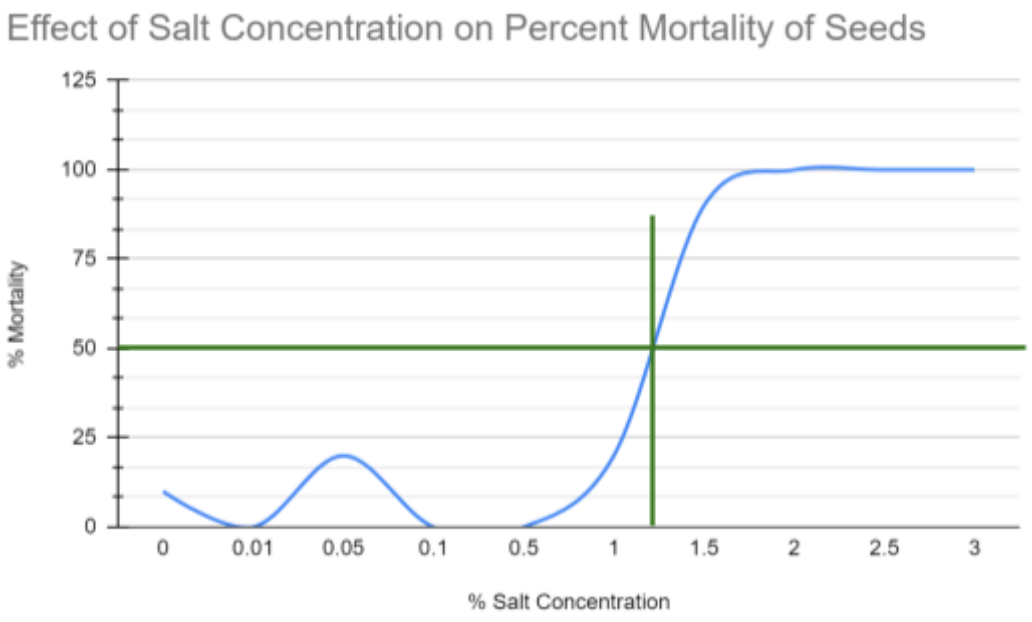
Average Root Growth

Concentration (% salt)	0	0.01	0.05	0.1	0.5	1	1.5	2	2.5	3
Average	20.50	35.20	17.50	33.10	20.40	7.30	1.50	0.00	0.00	0.00

2. Seeds that have germinated will have a sprout protruding from the seed; if there is no sprout, the seed has not germinated. Count and record the number of seeds that did not germinate.
3. Salt can also impact the quality of root development. Carefully lay each germinated seed next to a ruler and measure it from the seed to the end of the root (to the nearest millimeter). Calculate the average root length for each Petri dish and record it in the data table. Ungerminated seeds should be included in this calculation and are considered to have a root length of 0 mm.

ANALYSIS

3. Insert dose response curve from your data.



- Plot percent mortality versus salt concentration for each dilution.
- Connect the points with a smooth fit line. The data should fit a logistic (s-shaped) curve.

4. Identify the independent variable. On which axis is it plotted?

The independent variable is the % Salt Concentration which is plotted on the x-axis.

5. Identify the dependent variable. On what axis is it plotted?

The dependent variable is the % Mortality which is plotted on the y-axis.

6. Identify and label the LD_{50} for radish seeds on the graph.
   - To determine the LD_{50} draw a horizontal line across from 50 percent mortality on the y-axis to where it intersects the smooth-fit line.
   - Draw a vertical line down to where it intersects the x-axis; this concentration is the LD_{50}.

DISCUSSION AND CONCLUSION

1. Describe how the control was used to validate the results of your experiment.

The control validated the results of the experiment because it proved that the salt concentration was the factor that killed the seeds.

2. Based on the data, make a claim about how increasing salt concentration affected the length of the roots that sprouted with each concentration.

Increasing the salt concentration caused the lengths of the roots to decrease.

3. Explain how the data illustrate that radish seeds have a *range of tolerance* for salinity.

The data illustrates that radish seeds have a range of tolerance for salinity because the only salinity levels where no seeds died were between but didn't include the max salinity tried and the minimum salinity tried.

4. Identify a modification you could make to the experiment and explain how that would alter the results.

I could change the experiment to have a higher sample size so instead of 10 seeds in each petri dish, having maybe 15 or 20 instead would increase the accuracy of the results.

Lethal dose 50% (LD_{50})
The dose or concentration of a substance that is lethal to 50 percent of the population of a particular species.

Dose-response curve:
A graph used to illustrate the relationship between exposure (the dose) to a particular drug, toxin, or pollutant and a measured effect (the response) on an organism.

Range of tolerance:
The optimal range of a specific factor or condition in which an organism can maintain homeostasis. Organisms have a range of tolerance for various pollutants. Outside of this range, organisms may experience physiological stress, limited growth, reduced reproduction, and in extreme cases, death.

CHECK FOR UNDERSTANDING

DEBRIEFING QUESTIONS

1. Identify the salt you would recommend for use in deicing roads. Support your claim with evidence.

There was only one type of salt we tested.

2. Propose an alternative strategy municipalities and homeowners could implement to manage icy roads. Include your reasoning.

Homeowners could use heated driveways to never have to salt their driveways. This solution would be better than salting because excess salting can cause damage to the ecosystem and harm the environment system.

3. Justify the solution you proposed by explaining another potential advantage.

A heated driveway will also remove the need for physical labor in the form of salting the driveway.

4. Identify a stakeholder, other than a municipal road crew or homeowner, that might want information on salt toxicity. Include your reasoning.

An environmental conservationist group might want information on salt toxicity as it could be a good case to prevent the use of salt on roads.

Designing an investigation and identifying the components of your research design is an important skill in AP Environmental Science. What variable are you manipulating? Have you included a control group that is not subjected to the treatment?